

# torch.linalg.ldl\_factor#

[https://pytorch.org/docs/stable/generated/torch.linalg.ldl\\_factor.html](https://pytorch.org/docs/stable/generated/torch.linalg.ldl_factor.html)

## Description:

Computes a compact representation of the LDL factorization of a Hermitian or symmetric (possibly indefinite) matrix. When  $A$  is complex valued it can be Hermitian (`hermitian= True`) or symmetric (`hermitian= False`). The factorization is of the form  $A = LDL^H$  (or  $A = LDL^T$  if `hermitian` is `False`).  $L$ ,  $D$ , and  $L^H$  (or  $L^T$ ) are stored in compact form in `LD`. They follow the format specified by LAPACK's `sytrf` function. These tensors may be used in `torch.linalg.ldl_solve()` to solve linear systems. Supports input of `float`, `double`, `complex` and `complexdouble` dtypes. Also supports batches of matrices, and if  $A$  is a batch of matrices then the output has the same batch dimensions.

# Function Signature

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```
torch.linalg.ldl_factor(A, *, hermitian=False, out=None)#
```

Parameters

Keyword Arguments

Returns

## Parameters

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### Keyword

`hermitian` (bool, optional) – whether to consider the input to be Hermitian or symmetric. For real-valued matrices, this switch has no effect. Default: False. `out` (tuple, optional) – tuple of two tensors to write the output to. Ignored if None. Default: None.

### Returns:

A named tuple (LD, pivots).

## Code Examples

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```
>>> A = torch.randn(3, 3)
>>> A = A @ A.mT # make symmetric
>>> A
tensor([[7.2079, 4.2414, 1.9428],
        [4.2414, 3.4554, 0.3264],
        [1.9428, 0.3264, 1.3823]])
>>> LD, pivots = torch.linalg.ldl_factor(A)
>>> LD
tensor([[ 7.2079,  0.0000,  0.0000],
        [ 0.5884,  0.9595,  0.0000],
        [ 0.2695, -0.8513,  0.1633]])
>>> pivots
tensor([1, 2, 3], dtype=torch.int32)
```

## Notes & Warnings

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### NOTE

Note When inputs are on a CUDA device, this function synchronizes that device with the CPU. For a version of this function that does not synchronize, see `torch.linalg.ldl_factor_ex()`.

## Detailed Documentation

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### Documentation

Computes a compact representation of the LDL factorization of a Hermitian or symmetric (possibly indefinite) matrix.

When  $A$  is complex valued it can be Hermitian (`hermitian= True`) or symmetric (`hermitian= False`).

The factorization is of the form  $A=LDL^TA = L D L^TA=LDLT$ . If `hermitian` is `True` then transpose operation is the conjugate transpose.

$LLL$  (or  $UUU$ ) and  $DDD$  are stored in compact form in  $LD$ . They follow the format specified by LAPACK's `sytrf` function. These tensors may be used in `torch.linalg.Ldl_solve()` to solve linear systems.

Supports input of `float`, `double`, `cfloat` and `cdouble` dtypes. Also supports batches of matrices, and if  $A$  is a batch of matrices then the output has the same batch dimensions.

When inputs are on a CUDA device, this function synchronizes that device with the CPU. For a version of this function that does not synchronize, see `torch.linalg.Ldl_factor_ex()`.

$A$  (Tensor) – tensor of shape  $(*, n, n)$  where  $*$  is zero or more batch dimensions consisting of symmetric or Hermitian matrices.

`hermitian` (bool, optional) – whether to consider the input to be Hermitian or symmetric. For real-valued matrices, this switch has no effect. Default: `False`.

`out` (tuple, optional) – tuple of two tensors to write the output to. Ignored if `None`. Default: `None`.

A named tuple ( $LD$ , `pivots`).

